

厦门大学

Modeling Music Influence

Integrating Social Network Analysis with Maximum Likelihood Estimation

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1 Introduction

1.1 Background

Music is a cultural product that people use to express their emotions and relax their bodies and minds. The influence of musicians on each other has a very important impact on the development and evolution of music. The development of computer technology allows us unprecedented access to data and computing power to analyze the network of influences between musicians, from which we can extract and analyze the key factors that influence the process of musical development, which will also help us understand the meaning of music better.

1.2 Problem Statement

In this problem, I need to accomplish the following objectives:
Search data and use the data to create a (multiple) directed network(s) of musical influence, where influencers are connected to followers. Develop parameters that capture "music influence" in this network. Explore a subset of musical influence by creating a subnetwork of your directed influencer network. Describe this subnetwork. Reveal what my "music influence" measures in this subnetwork.

2 Preparation

2.1 Data Search

In the pursuit of comprehensive and relevant data for our research on musical influence and the dynamics between artists, a meticulous data - sourcing process was undertaken. Data collection is a fundamental step that can significantly impact the validity and reliability of research findings.

The "influence_data" dataset, which forms the cornerstone of our investigation, was sourced from AllMusic.com through a scraping process. This dataset is a rich repository of information, offering unique insights into the world of music over the past 90 years. It encompasses data on 5,854 artists, documenting the relationships between musical influencers and their followers. The data is self - reported by the artists themselves and also incorporates the opinions of industry experts, lending a multi - faceted perspective to the study.

Examining the structure of the "influence_data" dataset, it consists of several key columns that are essential for understanding the musical influence landscape. One of the columns likely contains the identification information of the influencers, which could be used to track and analyze the impact of well - known musicians over time. Another column is presumably dedicated to the identification of the followers, allowing us to map out the spread of influence across different artists. Additionally, there may be columns indicating the time period when the influence occurred, which is crucial for studying trends in musical influence over the 90 - year span covered by the dataset. There could also be columns related to the genre of music associated with both the influencers and followers, enabling us to explore how different musical genres have influenced one another.
Overall, the careful sourcing and utilization of the "influence_data" dataset position our research to make significant contributions to the field of musicology and related disciplines.

2.2 Assumption

We made the following assumptions to help us with our modeling. These assumptions are the promise of our subsequent analysis.

• Assuming that the data is accurate and there is no falsified data: This means our analysis accords with the truth.

• Assuming that followers can only be influenced by songs released before the current year, and that songs after the current year have no effect on the follower's for that year: this is done to measure the true influenceability of the influencers in a sliding window model.

• Assuming that the decade when the musician begin his carrer is the year when they debut, because the accurate years when the musicians debut are not given.

• Assuming that the musician's genre is limited to the genre provided by the data, and have no genre changes, that all his/her songs are in the style of his or her genre

2.3 Notation

The primary notations used in this paper are listed in Table 1.

Table 1: Parameter Settings

parameter	description
T	time-span influence factor
D_{Ci}	degree centrality
E_{Ci}	eigenvector centrality
MI	music influence

3 Methodology

3.1 Introduction of Social Network Analysis

Social Network Analysis (SNA) is a methodological framework used to examine the structure, dynamics, and patterns of relationships within a network of interconnected entities, such as individuals, organizations, or artifacts. Rooted in sociology and graph theory, SNA treats social systems as graphs, where nodes represent actors (e.g., people or groups) and edges denote the ties or interactions between them, such as friendships, collaborations, or influences.

At its core, SNA employs quantitative and qualitative techniques to measure network properties, including centrality (e.g., degree centrality to identify key influencers), density (the overall connectedness), and community detection (identifying clusters or subgroups). These metrics enable researchers to uncover hidden patterns, such as information flow, power dynamics, and diffusion processes.

To analyze the musical influences among musicians, I establish a directed network of musical influences with each musician as a node and directional influences among musicians as weights. Then I further explore the influence among musicians using social network analysis (SNA) to capture the *music influence* in the network to measure the influence of

each musician. Finally, we explore the sub-networks of the influence network and explore the significance of *music influence* in the sub-networks.

3.2 Social Network Analysis of Musical Influence Dynamics

3.2.1 Establishment of Social Network Analysis and Analysis Using SNA

▸ Establishment of Musical Influence Network

Directed graphs provide an effective framework for modeling the mutual influence relationships among musicians. Drawing from the *influencer_follower* relationships outlined in the *influence_data* dataset, we construct a directed network of musical influence, wherein each musician is represented as a node and the directed influences between them are encoded as weighted edges.

Data exploration reveals that these influences bifurcate into two main categories: intra-genre influences (within the same musical genre) and inter-genre influences (across different genres). Notably, influence flows bidirectionally across generations, with predecessors potentially influencing successors and vice versa. Thus, we classify inter-musician influences as follows.

From genre span: Influence occurs within the same genre or between different genres.

From time span: Junior influences predecessor or predecessor influences junior.

Fully considering the difference between the span of genres and the span of time in which the influence occurs, we define the weight of each side of the music influence network:

• **Time span impact factor**

Assuming that existing musician i influences musician j, the time difference Δt is defined as follows:

$$\Delta t = t_i - t_j \tag{1}$$

where t_i and t_j are the debut times of musician i and musician j, respectively. In general, juniors are more likely to be influenced by predecessors as they grow up, while predecessors are less likely to be influenced by juniors. Therefore, we can conclude that the probability of the older generation being influenced by the younger generation is lower. And if this happens, it means the influence of the younger generation is very strong. Compared to predecessors influencing juniors, juniors influencing predecessors should be considered that the juniors have a stronger influence. We therefore define the time-span influence factor T as follows:

$$T = \begin{cases} -\beta \Delta t, & \text{if } \Delta t < 0 \\ \\ \Delta t, & \text{if } \Delta t > 0 \end{cases} \tag{2}$$

• **Genre span impact factor**

In general, the mutual influences among musicians can be classified into two primary categories: cross-genre influences (those occurring within the same musical genre) and inter-genre influences (those spanning different genres). Within the same genre, musicians are more readily able to influence one another due to shared musical characteristics, such as stylistic similarities, thematic elements, and cultural contexts, which facilitate easier collaboration and idea exchange. In contrast, cross-genre influences are typically more

challenging to achieve, as they require overcoming barriers related to divergent aesthetics, traditions, and audience expectations. Nevertheless, when cross-genre influences do occur, they often result in greater overall impact, fostering innovation, genre fusion, and broader cultural shifts. Consequently, it can be posited that influencers who operate across genres exert a more substantial influence than those limited to a single genre, as evidenced by historical examples such as the cross-pollination between jazz and rock in the mid-20th century. This perspective underscores the importance of directed networks in capturing these dynamics, where edge weights could be adjusted to reflect not only the frequency but also the transformative potential of influences. The genre-spanning influence factor x_{ij} between influencer i and follower j is given as follows:

$$x_{ij} = \begin{cases} 1, & \text{cross-genre} \\ \\ 0, & \text{inter-genre} \end{cases} \tag{3}$$

• **Edge weights of a directed graph** Combining the above analysis on the time span and genre span impact factors, we give the formula for the directed graph weights w_{ij} as follows:

$$w_{ij} = (1 + x_{ij}) \cdot e^{\frac{T}{80}} \tag{4}$$

In equation(4), w_{ij} is the weight of the directed edge between musician nodes i to j. The time interval $|\Delta t|$ between the influencer and the influenced is at most 80, so T is normalized using $\frac{T}{80}$. w_{ij} also represents the intensity of the influence of musician i on musician j. The larger the value of w_{ij}, the stronger the intensity of musician i's influence.

▸ SNA Network Analysis Method

In order to capture the *musicl influence* in the established musical influence directed network, we analyze the directed network usinag social network analysis (SNA) method[3]. The influence of each musician is measured and analyzed by calculating the Degree Centrality of each musician in the musical influence directed network and further calculating the Eigenvector Centrality modified by the PageRank idea. The specific steps are as follows:

***Step 1*: Calculating Degree Centrality**

In an undirected network, the degree of a node can be used to measure the centrality of that point. Analogously, in the music influence directed network we built, we can use the sum of the out-degree weights of a node to measure the importance of a node's influence at the microscopic level. The assumption behind this metric is that nodes with many connections are important nodes, which means that musicians who have an impact on more people have a stronger influence. We therefore give the definition of degree centrality (D_{Ci}) for point i as shown in the equation(5)

$$D_{Ci} = \sum_{j=1}^{n} w_{ij} d_{ij} \tag{5}$$

In equation(5) n represents the total number of nodes in the network and d_{ij} denotes the existence of directed edges between node i to node j. And d_{ij} is defined as follows:

$$d_{ij} = \begin{cases} 1, & \text{no directed edge } from\ i\ to\ j \\ 0, & \text{there is directed edge } from\ i\ to\ j \end{cases} \tag{6}$$

***Step 2*: Calculating Eigenvector Centrality**

The importance of a node depends both on the number of its neighboring nodes and on the importance of its neighboring nodes. This means that the more important the nodes connected to node i, the more important node i will be. We define the eigenvector centrality E_{Ci} to measure the importance of node i at the macro level, as shown in the iterative equation(7)

$$\textit{iterative equation:} \quad E_{Ci} = \lambda \sum_{j=1}^{n} d_{ij} E_{Cj} \tag{7}$$

where *lambda* is a scaling constant and the initial value of the iteration is set in the equation(8)

$$\textit{initial value:} \quad E_{Cj} = D_{Cj}, \quad j = 1 \sim n \tag{8}$$

However, for the directed network we build, the above analysis process only takes the importance of neighboring nodes into account. To increase the influence of the nodes themselves on the outside, we modify the equation based on the PageRank idea.

PageRank[4] is an algorithm proposed by Sergey Brin and Lawrence Page to rank the importance of nodes in a network. The PageRank algorithm assumes that a node exerts influence on the system is the result of its own influence combining with the results of its followers' influences. Therefore, we modify the model shown in the following equation: Therefore, we modify the model shown in the following equation:

$$\textit{iterative equation:} \quad E_{Ci} = \lambda \sum_{j=1}^{n} d_{ij} E_{Cj} + (1-\lambda) E_{Ci} \tag{9}$$

After several iterations, E_{Ci} converges to a fixed value E'_{Ci}. We define the *musical influence* (MI_i) of musician i as follows:

$$MI_i = \frac{E'_{Ci}}{max\{E'_{Cj},\ j = 1, 2, \cdots, n\}} \tag{10}$$

3.2.2 Model Test on Musical Influence Network

Using all data given in the *influence_data*, the *musical influence* (MI) of each musician is calculated, and the MI of somes musician are shown in the Table(2).

Table 2: MI of Musicians

Musician	MI	**Musician**	MI
The Beatles	1	**Bob Dylan**	0.738
The Rolling Stones	0.546	**Chuck Berry**	0.5
Hank Williams	0.424	**Elvis Presley**	0.424
Jimi Hendrix	0.397	**Little Richard**	0.39
James Brown	0.387	**The Velvet Underground**	0.356

As can be seen from the above table, The Beatles has the highest MI, followed by Bob Dylan, The Rolling Stones, Chuck Berry, Elvis Presley and so on. The similarity between the most influential musicians we analyzed and *The 20 Most Representative Musicians in the World Jointly* selected by CNN and Songlines (a world-class music publication) is relatively high. We can conclude that the establishment and analysis process of the model is accurate, reasonable and feasible.

4 Analysis of The Result

Most real-world networks are not random networks, and a few nodes tend to have a large number of connections, while most nodes have only a small number of connections, generally obeying the Pareto Principle law[2]. Complex networks with degree distribution conforming to power-law distribution are usually called scale-free networks[1].

In section 3.2.3, we calculated the *musical influence* (MI) of each musician. A line graph of the frequency distribution of MI is plotted as shown in the blue solid line in the Figure(1).

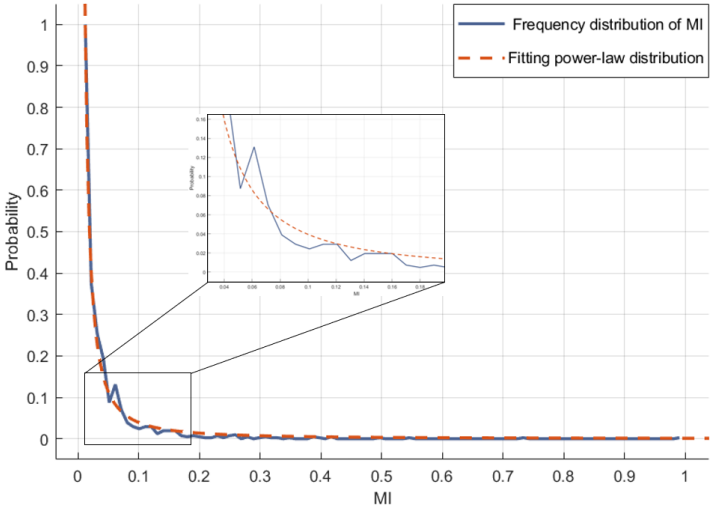


Figure 1: The Frequency Distribution of MI

To investigate the nature of the MI distribution, we fit the frequency distribution curve of MI. Observe that the distribution of MI in the figure resembles a power-law distribution, so the complementary cumulative distribution function of the power-law distribution is defined as equation(11)

$$P(X|X > x) = cx^{-a}, \qquad c > 0, a > 0 \tag{11}$$

We solve for the power-law distribution parameters using the maximum likelihood estimation, and the distribution parameters at a confidence level of 95% are shown in the Table(3).

Table 3: Distribution Parameters

distribution parameter	**estimated value**	**95% Confidence interval**
c	0.001173	(0.0009809, 0.001366)
α	-1.526	(-1.565, -1.488)

With an R^2 goodness-of-fit value of 0.9931, the Musical Influence Network can be considered a scale-free network. The Musical Influence Network adheres to the Pareto Principle, meaning that only a small proportion of musicians exert influence on numerous other musicians, whereas the majority of musicians do not influence others. This indicates that a small number of highly influential musicians play a crucial role in shaping the music world, and this phenomenon will be analyzed in detail in the subsequent sections.

5 Analysis of Musical Influence Sub-network

In the model construction phase, we partitioned the time span and genre span into four categories: within-genre influence, cross-genre influence, influence of predecessors on successors, and influence of successors on predecessors. We constructed separate independent sub-networks based on genre and time span, and employed Social Network Analysis (SNA) to investigate each musician's influence on predecessors (EI_{pre}) and successors (EI_{jun}), as well as influence within (EI_{wit}) and across (EI_{out}) genres. Notably, EI_{pre}, EI_{jun}, EI_{wit}, and EI_{out} constitute subsets of the total influence measure, EI.

To establish the musical influence sub-networks, we selectively extracted cross-genre influence data, within-genre influence data, predecessor influence data, and successor influence data from the *influence_data* dataset. Subsequently, we computed and normalized each type of musical influence separately. The calculated results for each influence type, along with the constructed musical influence sub-networks, are illustrated in Figure (2).

As depicted in Figure (2), the size of nodes in each sub-network corresponds to the Musical Influence (MI) value of the respective type of singer, while the thickness of the directed curves represents the influence weight. The figure reveals that within-genre influence occurs more frequently than cross-genre influence, and the influence of predecessors on successors is more prevalent than the reverse. This indicates that during musical interactions, musicians tend to influence those within the same genre more readily, and are more likely to exert influence on younger generations rather than their predecessors. Additionally, analysis of the MI sub-series highlights that The Beatles and Bob Dylan wield substantial influence over musicians both within and across genres. Notably, Bob Dylan demonstrates an exceptional ability, as he also exerts a strong influence on his predecessors.

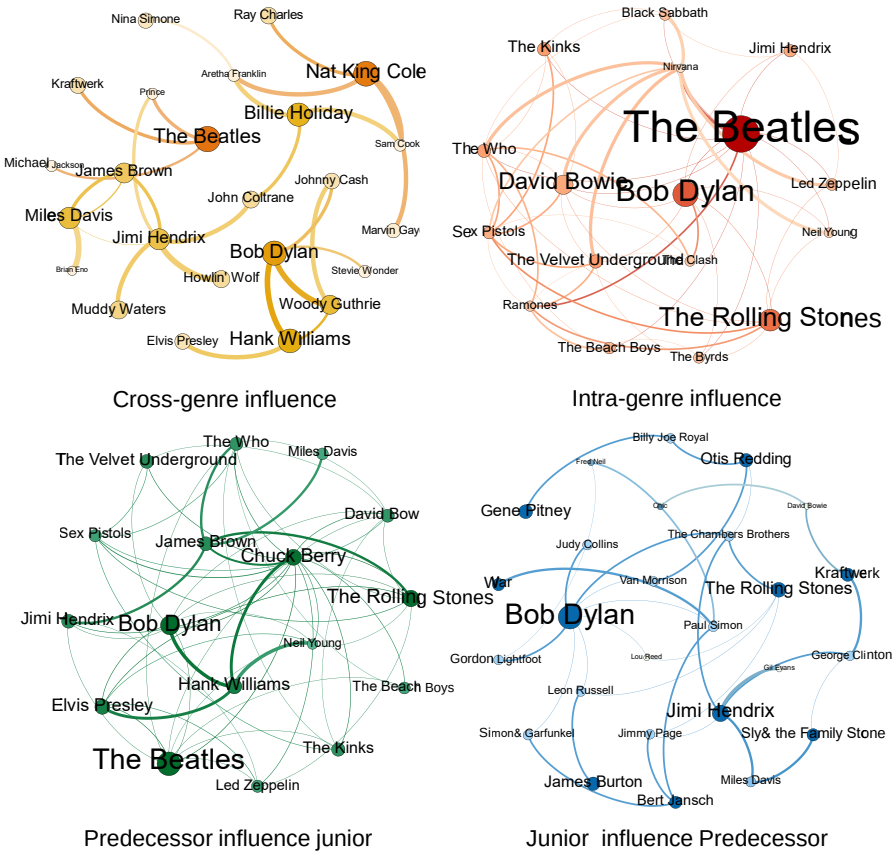


Figure 2: Sub-network of Musical Influence

6 Conclusion

This study employs social network analysis (SNA) to analyze musical influence networks, integrating temporal and genre dimensions to quantify *musical influence* (MI) across 5,854 artists. Key findings include:

1. **Scale-Free Network Structure**

The MI distribution follows a power-law pattern ($R^2 = 0.9931$), confirming the network is scale-free:

- A tiny subset of musicians (e.g., The Beatles, Bob Dylan) drives most influence, aligning with the Pareto Principle.
- Most artists exhibit minimal influence, typical of complex real-world networks.

2. **Temporal and Genre Asymmetry**

- Temporal Flow: Predecessors influence successors far more frequently than the reverse, though exceptions (e.g., Bob Dylan) highlight transformative reverse influence.
- Genre Dynamics: Within-genre influence is more common, but cross-genre influence carries higher weight, signaling its role in innovation (e.g., genre fusion).

3. **Methodological Validation**

Combining degree centrality (micro-level) and eigenvector centrality (macro-level, PageRank-modified) effectively identifies key influencers, consistent with industry rankings (e.g., CNN/Songlines).

4. **Implications**

- Theoretical: Extends SNA to musicology, linking influence patterns to contextual factors (time, genre).
- Practical: Informs music recommendation systems and cultural policy; highlights cross-genre innovation potential.
- Future Work: Refine models with dynamic genre/temporal data; explore subgenres/regional scenes.

5. **Summary**

Musical influence networks are scale-free, dominated by hub artists. Temporal precedence and genre affinity shape influence flows, while rare cross-genre/reverse influences drive evolution. This study provides a robust framework for decoding artistic diffusion dynamics.

7 Reflections and Insights

Throughout the process of writing this thesis, I have gained substantial knowledge and practical skills that have significantly enriched my academic experience. A major area of learning was social network analysis, which provided me with a comprehensive understanding of the structural and dynamic properties of networks. This analytical framework enabled me to explore complex relationships and interactions within social systems, deepening my insight into the subject matter.

In addition to theoretical learning, the thesis work greatly enhanced my programming skills. Implementing social network analysis algorithms required proficiency in coding, which I developed through hands-on practice. This experience not only improved my ability to write efficient and robust code but also strengthened my problem-solving capabilities in computational contexts.

Moreover, I became proficient in using LaTeX for document preparation. Mastering LaTeX allowed me to present my research in a professional and organized manner, especially when dealing with complex mathematical expressions and formatting requirements. This skill will undoubtedly be valuable for my future academic writing.

From a statistical perspective, I applied concepts learned in my mathematical statistics course, particularly Maximum Likelihood Estimation (MLE) and confidence intervals. Utilizing MLE enabled me to estimate model parameters effectively, while confidence intervals provided a rigorous way to quantify the uncertainty of these estimates. These statistical tools were crucial in validating the results and ensuring the reliability of my findings.

Overall, this thesis journey has been highly rewarding. It not only expanded my theoretical knowledge but also honed my technical skills and deepened my appreciation for rigorous academic research. I am confident that the competencies I have developed will serve as a strong foundation for my future studies and professional endeavors.

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